MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Sylvia Heaton, Permits Section, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: January 30, 2018

SUBJECT: Cadillac Waste Water Treatment Plant (WWTP)
Permit Review Biologist Recommendations
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0023531

Pursuant to your request, we have reviewed the NPDES permit reissuance application from the City of Cadillac for the Cadillac WWTP located in Cadillac, Michigan. The facility is requesting to continuously discharge a maximum of 3.2 million gallons per day (MGD) of treated municipal wastewater and discharge it through Monitoring Point 001A through Outfall 001 into the Clam River. The Clam River has a 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.9, 14, and 3.6 cubic feet per second (cfs), respectively.

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit application, the current NPDES permit, the Permits Section Toxics facility file, discharge monitoring report data from June 2013 through October 2017 as this is final effluent data since the previous WQBEL review, a Point Source Monitoring Survey conducted on October 14, 2014, and United States Geological Survey (USGS) topographical maps. Based upon the review of this information, we have the following recommendations:

Recommendations

1) The 2016 Water Quality and Pollution Control in Michigan Sections 303(d), 305(b), and 314 Integrated Report indicates the Clam River (Assessment Unit ID (AUID) 040601020303-01) is Not Supporting Water Quality Standards (WQS) (Category 4a Waterbody) for the Fish Consumption designated use due to polychlorinated biphenyls (PCBs) in the water column. All other designated uses are listed as Fully Supporting or Not Assessed. A statewide Total Maximum Daily Load (TMDL) for PCBs was approved by the United States Environmental Protection Agency (USEPA) on September 26, 2017. Several of the downstream AUIDs also have similar nonattainment issues.

Outfall 001

2) The current NPDES permit requires daily flow reporting. The Outfall 001 final effluent flow from June 2013 through October 2017 ranged from 1.14 to 3.61 MGD, averaging 2.1 MGD as compared to the design flow of 3.2 MGD. There was one day during this time period when the final effluent flow was greater than 3.2 MGD. Based on this information, we recommend the daily monitoring requirements for flow to be retained in the draft NPDES permit.

3) The current NPDES permit includes daily minimum and daily maximum final effluent limitations for pH of 6.5 and 9.0 Standard Units (S.U.), respectively. Compliance with these limits is measured as daily grab samples. Levels of pH in the Outfall 001 final effluent from June 2013 through October 2017 ranged from 6.3 to 8.1 S.U., averaging 7.2 S.U. There were three days when the daily minimum pH limit of 6.5 S.U. was exceeded. Based on this information, we recommend the draft permit retain the current final effluent limits and monitoring requirements for pH.

4) The current NPDES permit includes a monthly average total phosphorus concentration limit of 0.5 milligram per liter (mg/l) and a loading limit of 13 pounds per day (lbs/day) from May 1 through March 31 and a monthly average concentration limit of 0.65 mg/l and a loading limit of 17 lbs/day from April 1 through April 30. Compliance is determined as 24-hour composite samples collected five times per week. Concentrations and loading of total phosphorus in the Outfall 001 final effluent from June 2013 through October 2017 ranged from 0.03 to 4.2 mg/l, averaging 0.2 mg/l as a concentration and 0.48 to 107 lbs/day, averaging 4.9 lbs/day as a loading. The monthly average concentration was exceeded one time during this time period in October 2014. The load limit was never exceeded during this time period. To protect the Clam River and downstream surface waters from excessive nutrients, we recommend the draft NPDES permit retain the current limitations for total phosphorus.

5) The current NPDES permit does not include any monitoring requirements or final effluent limitations for total residual chlorine. In fact, the current NPDES permit indicates ultraviolet disinfection is used. The flow diagram and the brochure describing the history and treatment provided in the NPDES permit application indicates chlorine is used for disinfection. MDEQ District staff confirmed that ultraviolet disinfection is used and that no chlorine is used and instructed the Cadillac WWTP to update their flow diagram and facility brochure. Based on this information, we have no recommendations for total residual chlorine as it is understood ultraviolet disinfection is used. However, if chlorine is used, we recommend the draft NPDES permit to include a daily maximum WQBEL for total residual chlorine of 38 micrograms per liter (µg/l). If this limit is included, compliance with this limit shall be daily with samples collected as grab samples and analyzed using an approved USEPA method.

6) The current NPDES permit includes a monthly average WQBEL for available cyanide of 5.9 µg/l (0.16 lbs/day) with compliance determined as monthly grab samples. Concentrations of available cyanide in the Outfall 001 final effluent from June 2013 through October 2017 ranged from not detected (<2.0 ug/l) to 5.0 ug/l. These data indicate there is not a reasonable potential for available cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations and monitoring requirements for available cyanide.

7) The current NPDES permit includes a quarterly monitoring requirement for total copper. Outfall 001 final effluent concentrations for total copper collected from June 2013 through October 2017 ranged from not detected (<50 ug/l) to 86 ug/l. It should be noted the permittee reports a quantification level of 50 ug/l for the analysis of total copper. According to information provided during the previous WQBEL review, the Cadillac WWTP has the capability to analyze copper using its Atomic Absorption machine (flame method) which can reach a detection limit of 50 ug/l. This analytical method and quantification level is not sufficiently sensitive for the characterization of total copper in the Outfall 001 final effluent as the WQBELs for total copper are less than 50 ug/l. Cadillac District staff confirmed the facility is reporting a quantification level of 50 ug/l and has instructed the facility to achieve a much lower quantification level. Since the final effluent data reported as not detected using a quantification level of 50 ug/l are not sufficiently sensitive, these data were not used in the reasonable potential analysis. Based on this information, there is a reasonable potential for total copper to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average and daily maximum WQBELs of 11 ug/l (0.3 lbs/day) and 31 ug/l (0.82 lbs/day), respectively. Compliance with these limitations shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method found in 40 CFR Part 136 capable of achieving a quantification level of 1.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.

8) Final effluent data for total nickel ranged from not detected (<5.0 ug/l) to 85 ug/l. These data indicate there is a reasonable potential for total nickel to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for total nickel of 51 ug/l (1.4 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 5.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.

9) Final effluent data for total selenium ranged from not detected (<1.0 ug/l) to 6.6 ug/l. These data indicate there is a reasonable potential for total selenium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for total selenium of 5.7 ug/l (0.15 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 1.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.

10) Final effluent data for bis(2-ethylhexyl)phthalate ranged from not detected (<5.0 ug/l) to 19 ug/l. These data indicate there is a reasonable potential for bis(2-ethylhexyl)phthalate to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL for bis(2-ethylhexyl)phthalate of 31 ug/l (0.82 lbs/day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA method capable of achieving a quantification level of 5.0 ug/l. Appropriate language shall also be included in the draft NPDES permit to allow a monitoring frequency reduction from weekly to no less than quarterly after one year of testing has been completed.

11) The current NPDES permit includes a 12-month rolling average level currently achievable (LCA) final effluent limit for total mercury of 2.0 nanograms per liter (ng/l) (0.000053 lbs/day) with compliance samples collected as quarterly grab samples. Part I.A.3 of the current NPDES permit also describes the requirements for the Pollutant Minimization Program (PMP) for total mercury. Concentrations of total mercury in the Outfall 001 final effluent from June 2013 through October 2017 ranged from not detected (<0.5 ng/l) to 1.1 ng/l. Based on the final effluent data for total mercury, there is a reasonable potential for total mercury to be discharged from Outfall 001 at levels exceeding Michigan WQS. We recommend the draft permit retain the current LCA final effluent limit for total mercury of 2.0 ng/l (0.000053 lbs/day). Compliance with this limit should continue on a quarterly basis with samples collected as grab samples using USEPA method 1669 and analyzed using USEPA method 1631 with a quantification level of 0.5 ng/l. In addition, we recommend the PMP requirements be retained in the draft permit.

12) The current NPDES permit includes a chronic Whole Effluent Toxicity (WET) final effluent limitation of 1.1 chronic toxic units (TUc) and monitoring for acute WET. Compliance with this limitation is conducted as quarterly 24-hour composite samples and WET testing using *Ceriodaphnia dubia* (*C. dubia*) as the test species using chronic toxicity testing methods. Chronic and acute WET testing using fathead minnow as the test species is required in Part I.A.2, Additional Monitoring Requirements. From June 2013 through October 2017, the Outfall 001 final effluent was never acute or chronically toxic to either test species as all test results report 0.0 acute toxic units (TUa) and 0.0 TUc. Based on this information, there is not a reasonable potential for the Outfall 001 final effluent to be acutely or chronically toxic. We recommend the draft NPDES permit remove the chronic toxicity limit and reporting of the acute toxicity results. We do

recommend that acute and chronic WET testing be included in the conditions in Part I.A.2, Additional Monitoring Requirements with testing conducted using both fathead minnow and *C. dubia* as the test species for a total of eight tests (four tests on each species) using chronic toxicity testing methods and samples collected as 24-hour composite samples.

13) The analytical data provided in the permit application for the pollutants listed in the table below were all analyzed using an analytical method and quantification level that was not sufficiently sensitive to determine if these pollutants have the reasonable potential to be discharged at levels below Michigan WQS. This occurred during three of the four final effluent scans collected as part of Part I.A.2, Additional Monitoring Requirements. Based on this information, we recommend the draft permit include a quarterly monitoring requirement for these pollutants. Samples for these pollutants shall be collected as grab samples and analyzed using the following USEPA methods and quantification levels in the table below. Appropriate language should also be included in the draft permit to allow the permittee to request a monitoring frequency after 12-months to reduce the monitoring frequency to no less than annually. Alternate methods may be requested as long as they are USEPA approved methods that can meet the quantification level specified.

Parameter	**CAS Number**	**EPA Method**	**Quantification Level (µg/L)**
3,3-Dichlorobenzidine	91941	605	1.5
Benzidine	92875	605	0.1
Hexachlorobenzene	118741	612	0.01
Hexachlorobutadiene	87683	612	0.01
Hexachlorocyclopentadiene	77474	612	0.01
Acrylonitrile	107131	EPA Approved	1.0
4-Chloro-3-methylphenol	59507	EPA Approved	5.0
Fluoranthene	206440	EPA Approved	1.0
Pentachlorophenol	87865	EPA Approved	1.8
Phenanthrene	85018	EPA Approved	1.0
2,4,6-Trichlorophenol	88062	EPA Approved	5.0

14) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

Please contact me at (517) 284-5599 if you should have any questions regarding the recommendations above.